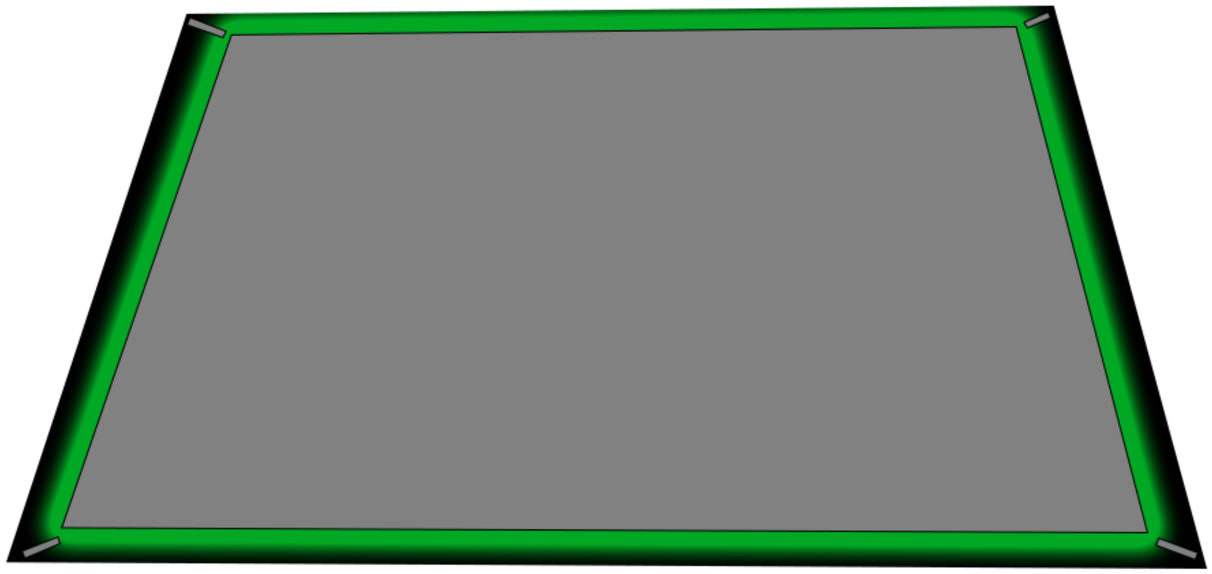
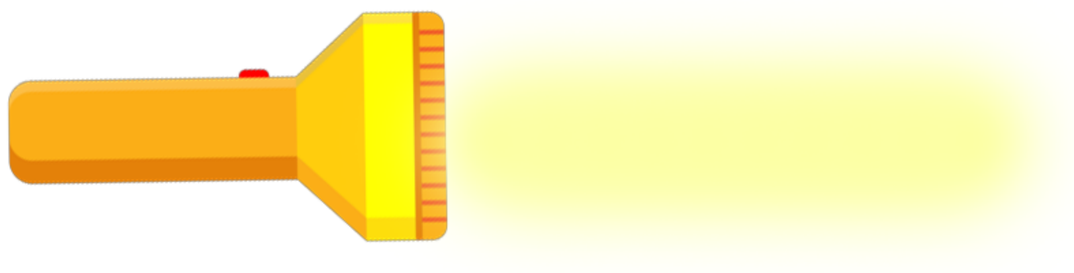
Pressure Plate:

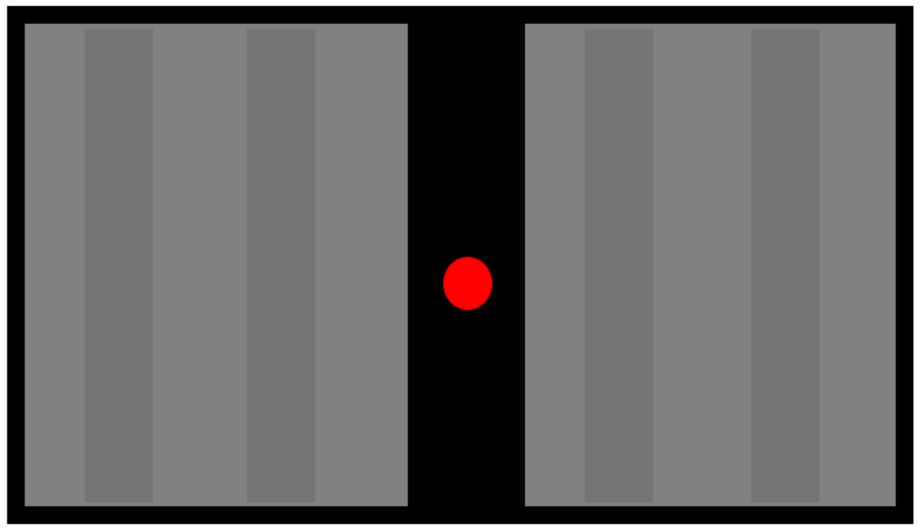


Properties:	-Activation threshold (numeric value) -Current weight (numerical value) -State (active/inactive) -Toggles between red and green (visual aid of active/inactive)
Behaviours:	-Activates only when the total weight on it (or exceeds) meets the threshold. -Deactivates immediately if weight drops below threshold
Relationships:	-Receives weight from weighted blocks, rolling boulders or players standing on it. -Controls connected barrier gates.
Feedback:	-Plate visibly lowers when activated -Glows or changes colour when activated.

Flashlight:



Light Barrier:



Properties:

- Beam direction (Based on where the player is facing)
- Beam range (fixed distance)
- Emission state (on/off)

Behaviours:

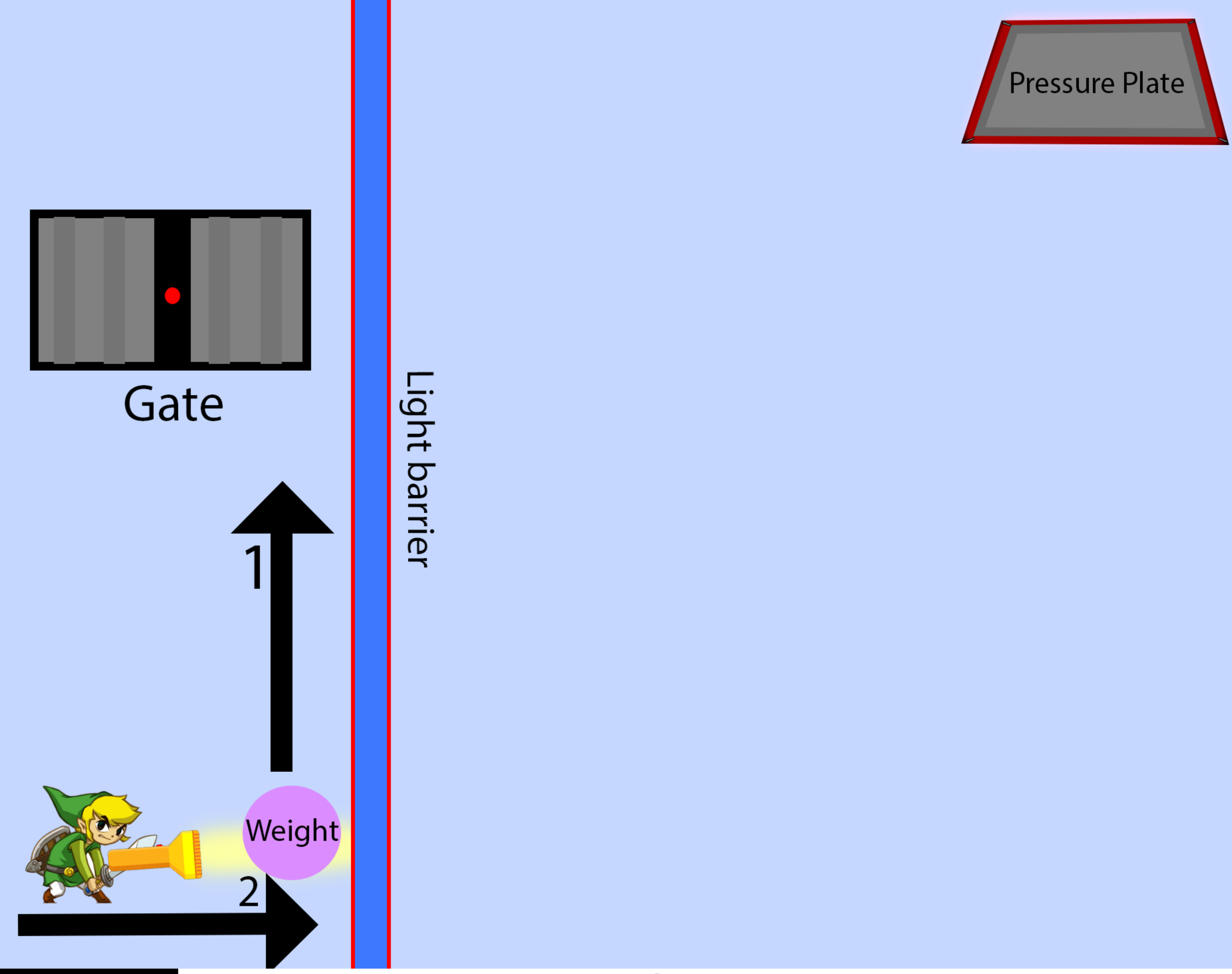
- Emits a beam only while activation button is held.
- Sends an illumination signal to light barriers.
- Stops sending a signal immediately after the beam is no longer making contact.

Relationships:

- Interacts directly with light barriers.
- Can be blocked by environmental objects (like weights).
- Requires player positioning and orientation to function properly

Feedback:

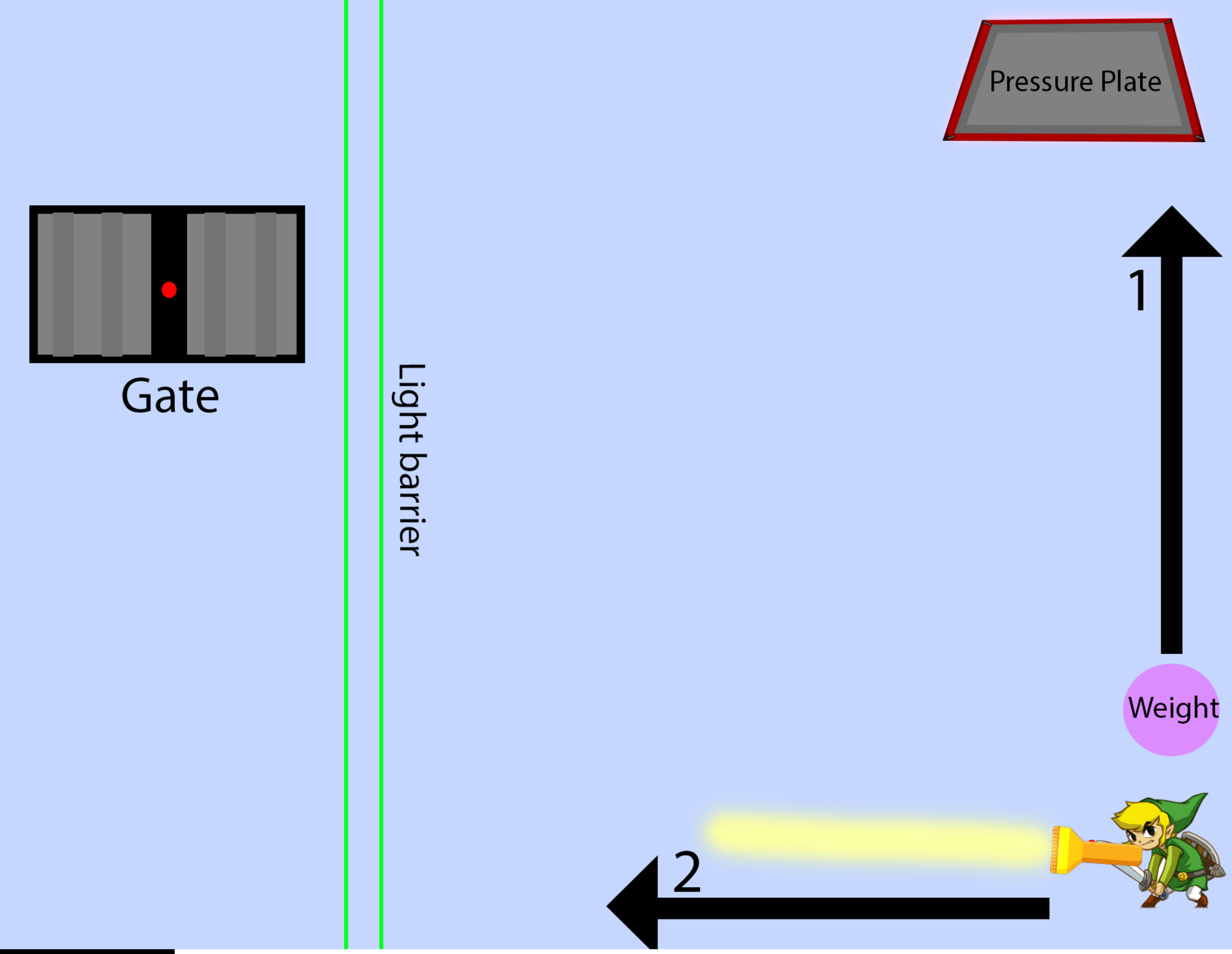
- Visible cone of light projecting from the player
- Subtle glow appears on light barrier when illuminated
- Hum and light activation feedback while beam is active.



A

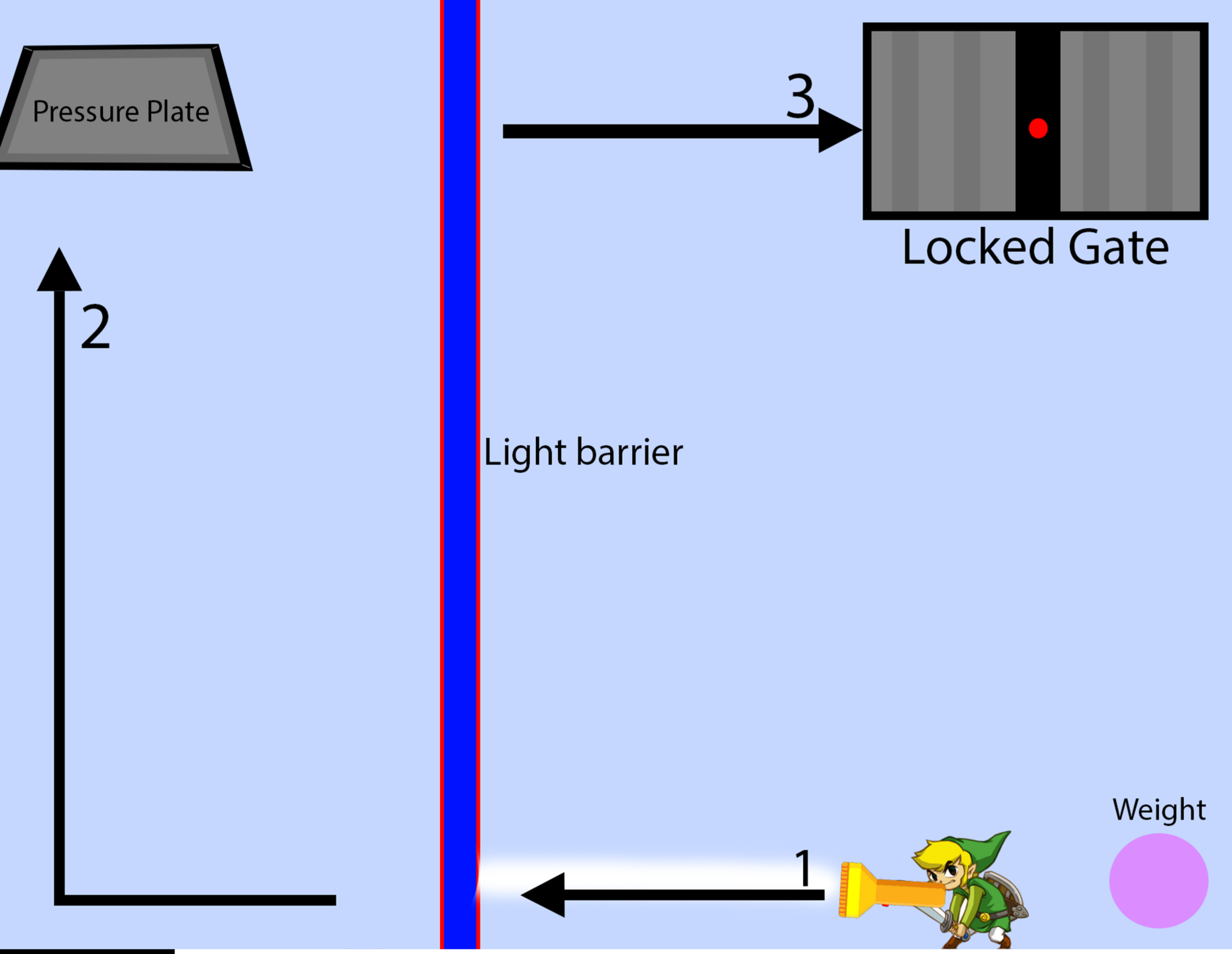
1: Move weight in order for beam to it light barrier

2: Hit wall with light to lower barrier



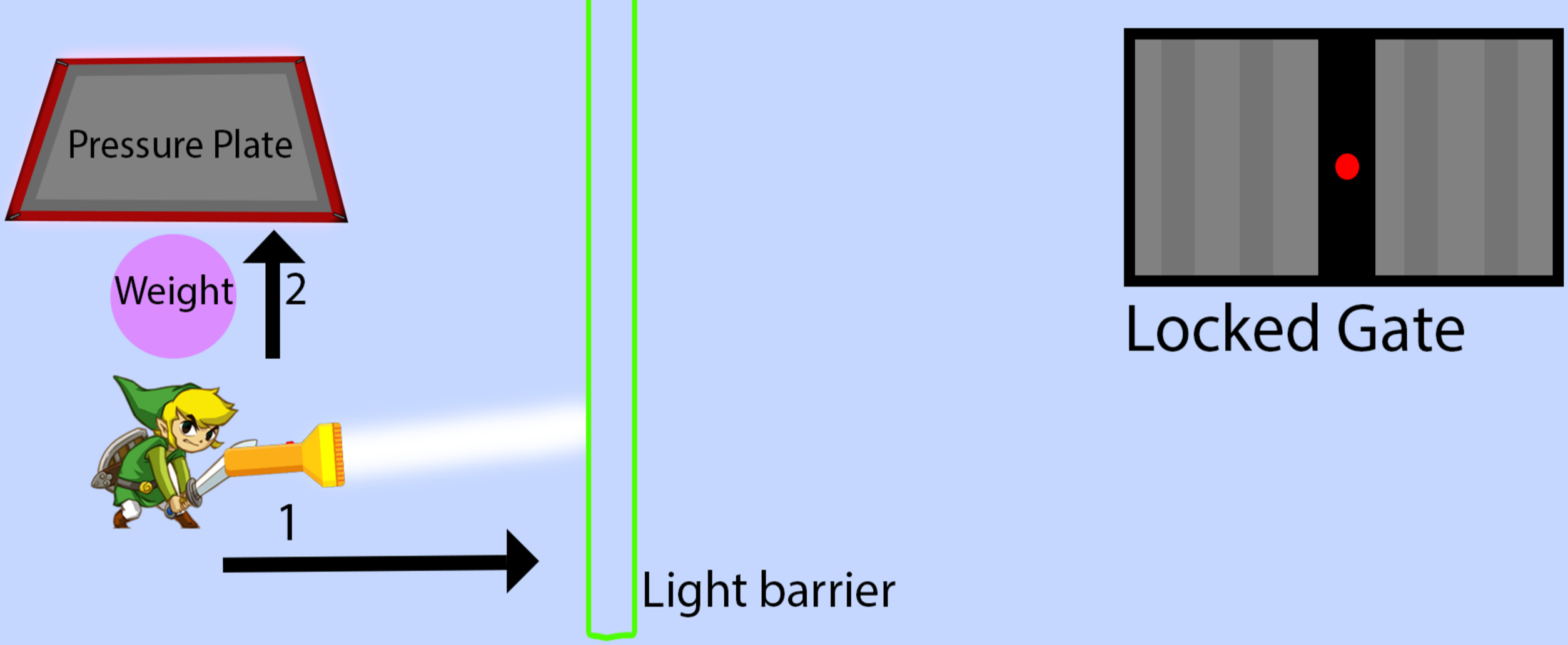
B

1: Move weight to pressure plate to open gate.
2: Flash beam at light barrier to open to reach gate



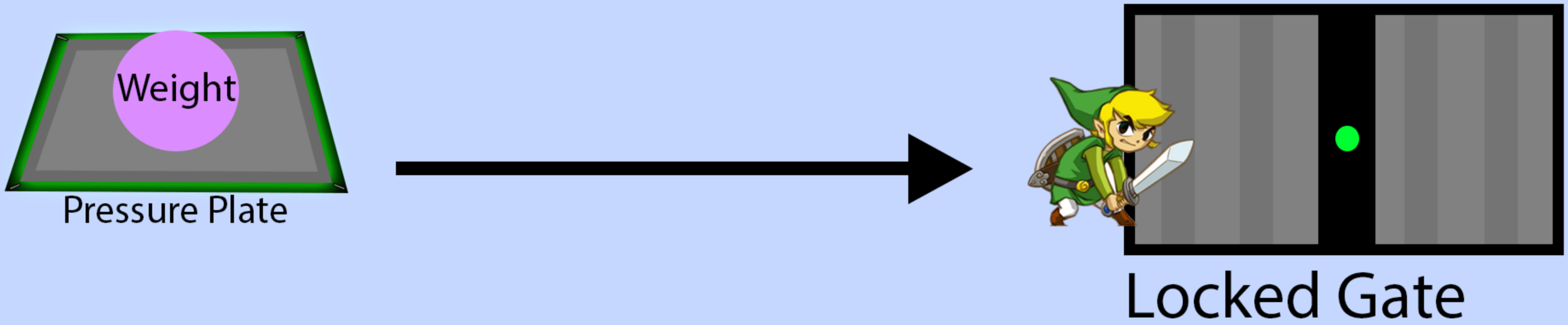
A

1: Hit light barrier with beam to drop wall
2: Move weight to activate pressure plate and open gate



B

1: Beam the light barrier to open.
2, Apply weight to pressure plate.



C

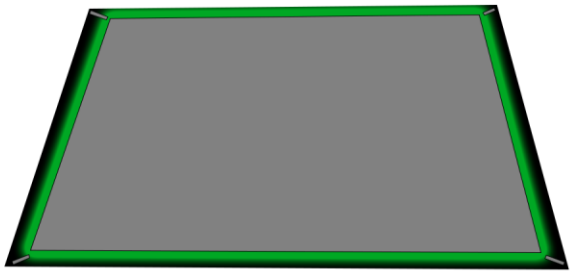
1: Escape the level

Assignment #2:

Systemic Games with Emergent Systems

Nick Frasiak

Weight and Pressure System

Pressure Plate:	
Properties:	-Activation threshold (numeric value) -Current weight (numerical value) -State (active/inactive) -Toggles between red and green (visual aid of active/inactive)
Behaviours:	-Activates only when the total weight on it (or exceeds) meets the threshold. -Deactivates immediately if weight drops below threshold
Relationships:	-Receives weight from weighted blocks, rolling boulders or players standing on it. -Controls connected barrier gates.
Feedback:	-Plate visibly lowers when activated -Glows or changes colour when activated.

Pressure Plate:

The pressure plate is a state-based environmental object that responds to weight rather than a toggle. Players combine blocks and moving objects to activate pressure plates and open barriers which create puzzles that depend on timing, position and combining objects. Unlike a simple toggle switch the plates state depends on the presence of weight making it more dynamic and dependent on the space around you.

The pressure points behaviour is conditional and continuous rather than instantaneous. When enough pressure or weight is applied, the plate will transition to the active state and emits a signal event to any connected object (such as a barrier). If the weight drops below the threshold, it immediately returns to inactive and sends a corresponding deactivation signal.

The pressure plate will contain the following key properties:

- Activation threshold (numeric value)
- Current weight (numerical value)
- State (active/inactive)
- Toggles between red and green (visual aid of active/inactive)

What is it?:

A plate on the floor that activates when enough weight is on top of it.

Behaviour:

- Activates only when the total weight on it (or exceeds) meets the threshold.
- Deactivates immediately if weight drops below threshold

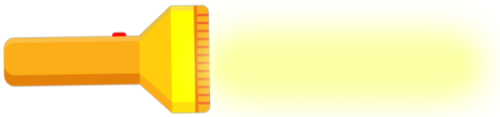
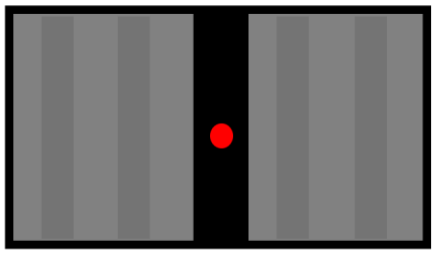
Relationships:

- Receives weight from weighted blocks, rolling boulders or players standing on it.
- Controls connected barrier gates.

Feedback:

- Plate visibly lowers when activated
- Glows or changes colour when activated.
- Click sounds or mechanical cue plays on activation.

Flashlight and Light Barrier

Flashlight:	Light Barrier:
	

Properties:	- Beam direction (Based on where the player is facing) - Beam range (fixed distance) - Emission state (on/off)
Behaviours:	- Emits a beam only while activation button is held. - Sends an illumination signal to light barriers. - Stops sending a signal immediately after the beam is no longer making contact.
Relationships:	- Interacts directly with light barriers. - Can be blocked by environmental objects (like weights). - Requires player positioning and orientation to function properly
Feedback:	- Visible cone of light projecting from the player - Subtle glow appears on light barrier when illuminated - Hum and light activation feedback while beam is active.

Flashlight:

The flashlight is a directional player ability that emits a focused beam of light used to interact with light sensitive environmental objects (Light Barrier). Instead of relying on weight like the pressure plate system, this system depends on the players orientation, the player must aim the beam at the light barrier to temporarily disable them, this creates traversal challenges based on timing and movement. Unlike a permanent state change the flashlight creates temporary openings that require quick decision making.

The flashlights behaviour is input based and continuous rather than automatic. When the player holds the flashlight button the beam will project in the direction the player is facing. If the beam makes contact with the light barrier, it sends a signal and the wall (barrier) drops. When the player releases the button or turns away the signal will stop and the barrier will re appear.

The pressure plate will contain the following key properties:

- Beam direction (Based on where the player is facing)
- Beam range (fixed distance)
- Emission state (on/off)

What is it?:

A player controlled light source that emits a directional beam to interact with the light barriers.

Behaviour:

- Emits a beam only while activation button is held.
- Sends an illumination signal to light barriers.
- Stops sending a signal immediately after the beam is no longer making contact.

Relationships:

- Interacts directly with light barriers.
- Can be blocked by environmental objects (like weights).
- Requires player positioning and orientation to function properly

Feedback:

- Visible cone of light projecting from the player
- Subtle glow appears on light barrier when illuminated
- Hum and light activation feedback while beam is active.